

Instructions

This mock mirrors the Tredence data-science assessment: a mix of aptitude, verbal, statistics, machine learning MCQs plus SQL and Python coding.

Section 1 - Aptitude & Reasoning (8 Q)

Section 2 - Verbal (4 Q)

Section 3 - Statistics & Probability (8 Q)

Section 4 - Machine Learning (8 Q)

Section 5 - SQL (2 problems)

Section 6 - Python (1 problem)

Attempt in one 2-hour sitting.

Section 1 - Aptitude & Reasoning

Q1. What is 12.5% expressed as a fraction?

(A) $\frac{1}{6}$

(B) $\frac{1}{8}$ ✓

(C) $\frac{1}{4}$

(D) $\frac{1}{5}$

Answer: (B) $\frac{1}{8}$

Q2. 250 is what percent of 1000?

(A) 20%

(B) 25% ✓

(C) 30%

(D) 15%

Answer: (B) 25%

Q3. If 60% of the students passed and 80 students failed, how many students are there in total?

(A) 160

(B) 200 ✓

(C) 180

(D) 240

Answer: (B) 200

Q4. A company's annual sales (in Rs. lakh) were: 2019=40, 2020=50, 2021=60, 2022=80. What is the total sales over these four years (in Rs. lakh)?

(A) 210

(B) 220

(C) 230 ✓

(D) 240

Answer: (C) 230

Q5. (Same data) What is the percentage increase in sales from 2019 to 2022?

(A) 50%

(B) 80%

(C) 100% ✓

(D) 120%

Answer: (C) 100%

Q6. Find the next term: 4, 9, 16, 25, ?

(A) 30

(B) 36 ✓

(C) 49

(D) 35

Answer: (B) 36

Q7. Find the next term: 2, 4, 8, 16, ?

(A) 24

(B) 32 ✓

(C) 20

(D) 30

Answer: (B) 32

Q8. Find the odd one out: 3, 5, 7, 9, 11.

(A) 5

(B) 7

(C) 9 ✓

(D) 11

Answer: (C) 9

Section 2 - Verbal

Q1. Choose the synonym of METICULOUS.

(A) Careless

(B) Careful ✓

(C) Quick

(D) Rough

Answer: (B) Careful

Q2. Choose the synonym of CONCISE.

(A) Lengthy

(B) Brief ✓

(C) Vague

(D) Complex

Answer: (B) Brief

Q3. Neither of the two answers ____ correct.

(A) are

(B) is ✓

(C) were

(D) have

Answer: (B) is

Q4. Choose the correct sentence.

(A) He is one of the best player.

(B) He is one of the best players. ✓

(C) He is one of best player.

(D) He is a best players.

Answer: (B) He is one of the best players.

Section 3 - Statistics & Probability

Q1. Find the mean of 2, 4, 6, 8, 10.

(A) 5

(B) 6 ✓

(C) 7

(D) 8

Answer: (B) 6

Q2. Find the median of 3, 1, 4, 1, 5, 9, 2.

(A) 2

(B) 3 ✓

(C) 4

(D) 1

Answer: (B) 3

Q3. Find the mode of 2, 3, 3, 5, 7, 3, 8.

(A) 2

(B) 3 ✓

(C) 5

(D) 7

Answer: (B) 3

Q4. A p-value less than the significance level (e.g. 0.05) suggests you should:

(A) Accept the null hypothesis

(B) Reject the null hypothesis ✓

(C) Ignore the result

(D) Increase the sample size

Answer: (B) Reject the null hypothesis

Q5. A Type I error is:

(A) Failing to reject a false null hypothesis

(B) Rejecting a true null hypothesis ✓

(C) Accepting the alternative correctly

(D) A sampling error

Answer: (B) Rejecting a true null hypothesis

Q6. A Type II error is:

(A) Rejecting a true null hypothesis

(B) Failing to reject a false null hypothesis ✓

(C) A measurement error

(D) A rounding error

Answer: (B) Failing to reject a false null hypothesis

Q7. The Central Limit Theorem states that the sampling distribution of the mean:

(A) Is always uniform

(B) Approaches a normal distribution as sample size increases ✓

(C) Has zero variance

(D) Equals the population distribution

Answer: (B) Approaches a normal distribution as sample size increases

Q8. For a normal distribution, approximately what percent of data lies within one standard deviation of the mean?

(A) 50%

(B) 68% ✓

(C) 95%

(D) 99.7%

Answer: (B) 68%

Section 4 - Machine Learning

Q1. Overfitting occurs when a model:

(A) Performs poorly on both training and test data

(B) Performs well on training data but poorly on unseen data ✓

(C) Has too few parameters

(D) Cannot be trained

Answer: (B) Performs well on training data but poorly on unseen data

Q2. Which technique helps reduce overfitting?

(A) Increasing model complexity

(B) Regularization ✓

(C) Removing the validation set

(D) Using fewer training epochs only

Answer: (B) Regularization

Q3. Supervised learning requires:

(A) Unlabeled data

(B) Labeled data ✓

(C) No data

(D) Only test data

Answer: (B) Labeled data

Q4. Which of the following is an unsupervised learning algorithm?

(A) Logistic Regression

(B) K-means clustering ✓

(C) Decision Tree classifier

(D) Linear Regression

Answer: (B) K-means clustering

Q5. Which metric is most appropriate for an imbalanced classification problem?

(A) Accuracy

(B) F1-score ✓

(C) Total count

(D) Learning rate

Answer: (B) F1-score

Q6. Precision is defined as:

(A) $TP / (TP + FN)$

(B) $TP / (TP + FP)$ ✓

(C) $TN / (TN + FP)$

(D) TP / Total

Answer: (B) $TP / (TP + FP)$

Q7. Recall (sensitivity) is defined as:

(A) $TP / (TP + FP)$

(B) $TP / (TP + FN)$ ✓

(C) $TN / (TN + FN)$

(D) $FP / (FP + TN)$

Answer: (B) $TP / (TP + FN)$

Q8. A confusion matrix shows:

(A) Feature correlations

(B) Counts of true/false positives and negatives ✓

(C) Model weights

(D) Learning curves

Answer: (B) Counts of true/false positives and negatives

Section 5 - SQL

Problem 1: Second Highest Salary

From the employee table, find the second highest distinct salary. If it does not exist, return NULL.

```
SELECT MAX(salary) AS second_highest
FROM employee
WHERE salary < (SELECT MAX(salary) FROM employee);
```

Expected Output

```
second_highest
60000
```

Problem 2: Count Employees per Department

Count the number of employees in each department and order the result by employee count descending, then department name ascending.

```
SELECT dept, COUNT(*) AS emp_count
FROM emp
GROUP BY dept
ORDER BY emp_count DESC, dept ASC;
```

Expected Output

```
dept|emp_count
Tech|3
Sales|2
HR|1
```

Section 6 - Python

Problem 1: Most Frequent Word

Difficulty: *Easy* | Pattern: *Hash Map / Counting*

Problem Statement

Given a sentence, return the word that appears most frequently. If several words tie, return the one that appears first. Comparison is case-insensitive.

Input Format

A single line: the sentence.

Output Format

The most frequent word (lowercase).

Constraints

Words are separated by single spaces.

Sample Input

```
the cat sat on the mat the cat
```

Sample Output

```
the
```

Python Solution

```
def most_frequent_word(sentence):
    counts = {}
    best_word, best_count = None, 0
```

```
for word in sentence.lower().split():
    counts[word] = counts.get(word, 0) + 1
    if counts[word] > best_count:
        best_count = counts[word]
        best_word = word
return best_word

print(most_frequent_word('the cat sat on the mat the cat'))
```

Step-by-Step Explanation

Count each word in a dictionary. Track the best word as you go; updating only when a strictly higher count appears preserves the first word on ties. 'the' occurs 3 times.

Dry Run

```
the->1 (best=the)
cat->1
sat->1
on->1
the->2 (best=the)
mat->1
the->3 (best=the)
cat->2
Result = the
```

Time Complexity: **O(N)** | Space Complexity: **O(K)** distinct words

Tredence

Full-Length Mock Test

Applied Role(s):

[Analyst - Data Scientist](#)

Compiled from verified previous-year patterns, community-reported questions and research-backed company-style practice.

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